Math 101 Final Exam — Form Mock

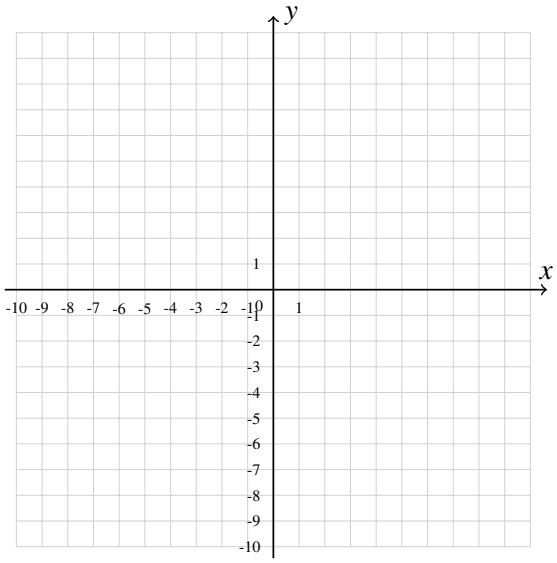
Name: _____ **Section:** ◯ 12 ◯ 29

Instructor: David Martinez **Total: 100 points**

Show all work. Each question is worth 5 points. Scientific calculators only. No graphing calculators, phones, or calculator sharing.

1. (5 pts) Graph $f(x) = x^2 - 6x + 8$. State the vertex and axis of symmetry.

Vertex: _____ Axis: _____

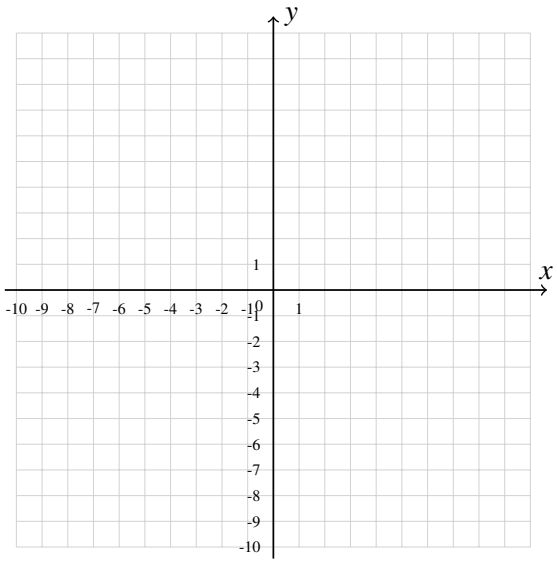


2. (5 pts) Let $f(x) = \sqrt{2x + 25}$ and $g(x) = x - 5$. Find and simplify $f(g(x))$ and $g(f(x))$.

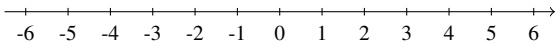
3. (5 pts) Evaluate $\log_{49}(7)$ without a calculator. Show how you know.

4. (5 pts) Find an equation of the line perpendicular to $3y + 9x = 6$ through (2,4).

5. (5 pts) Graph $f(x) = -2^x + 3$. Plot at least two points and identify the horizontal asymptote.



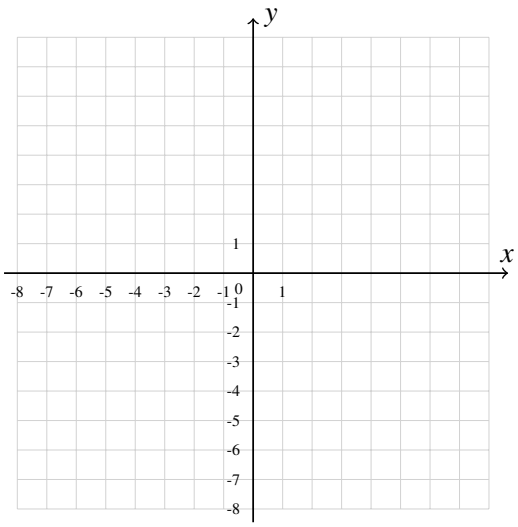
6. (5 pts) Solve the compound inequality, give interval notation, and graph the solution: $-5x + 1 \geq 16$ OR $3x - 2 > -2x + 8$.



7. (5 pts) Multiply and simplify: $\frac{x^2+5x+6}{x^2+4x+3} \cdot \frac{2x+2}{x^2+5x+6}$.

8. (5 pts) Divide and simplify in $a + bi$ form: $\frac{3+2i}{2-i}$.

9. (5 pts) Graph the piecewise function $f(x) = \begin{cases} x - 2, & x \leq 2 \\ -x + 4, & x > 2 \end{cases}$. Mark open and closed endpoints.

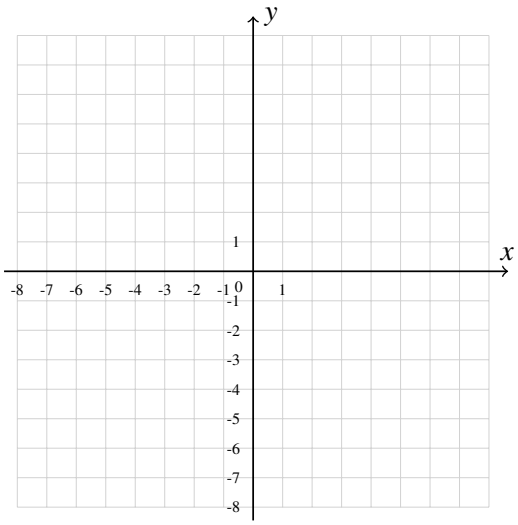


10. (5 pts) Rationalize the denominator and simplify: $\frac{2-\sqrt{3}}{3+\sqrt{2}}$.

11. (5 pts) Simplify with positive exponents only: $\left(\frac{x^2y^{-3}z}{x^{-2}yz^{-3}}\right)^{-2}$.

12. (5 pts) Solve the quadratic equation and simplify: $4x^2 = 8x + 3$.

13. (5 pts) Graph the system of inequalities, shading the solution region: $y > \frac{1}{2}x - 1$ and $y < -x + 2$.



14. (5 pts) Solve for x, checking excluded values: $\frac{2}{x-3} + \frac{x}{x-2} = \frac{-x+10}{x^2-5x+6}$.

15. (5 pts) A rocket has height $h(t) = -16t^2 + 96t + 64$ feet after t seconds. Find the maximum height and when it occurs.

16. (5 pts) Gas volume varies inversely with pressure. At 400 mmHg the volume is 3600 cm^3. Find the volume at 600 mmHg.

17. (5 pts) A plane flies 450 miles in the same time a car travels 150 miles. The car is 60 mph slower. Find the plane's speed.

18. (5 pts) How much 40% solution and 60% solution must be mixed to make 150 mL of a 48% solution? Set up an equation.

19. (5 pts) A 180-inch tube is cut into two pieces. One piece is 5 times the other. Find both lengths.

20. (5 pts) One worker finishes a job in 20 hours. Together with a second worker, the job takes 4 hours. How long would the second worker take alone?